



ROSHENI BOLONNE

THIRD-YEAR UNDERGRADUATE | INFORMATION TECHNOLOGY - AI SPECIALIZATION

PROFILE

Third-year IT undergraduate at SLIIT specializing in Artificial Intelligence (AI). Currently building skills in Python, machine learning, computer vision, NLP, and full-stack development with the MERN stack and learning best by building real projects with a pragmatic approach rather than just following tutorials. A quick and smart learner who enjoys figuring things out and isn't afraid to start something new. Seeking an internship in AI/ML, data science, or full-stack development where I can learn fast and grow alongside an experienced team.

Current GPA: 3.92/4.0

EDUCATION

BSc (Hons) in Information Technology - AI Specialization 2024-2028
Sri Lanka Institute of Information Technology | GPA: 3.92 / 4.0

Relevant Coursework: Introduction to Programming & OOP, Data Structures and Algorithms, Database Design and Development, Artificial Intelligence and Machine Learning, Probability & Statistics for Computing, Discrete Mathematics, Data Communication and Networking, Software Engineering, Web and Mobile Technologies

Activities and Societies: IEEE Society, Microsoft Club Alpha Member, SEDS Biotechnical Division Member

G.C.E. Advanced Level - Biological Science 2020-2023
St. Bridget's Convent, Colombo 07

O/L Results: 9As

A/L Results:

B: Chemistry

C: Physics

C: Biology

Activities and Societies: Science Society, Western Band, School Choir, Badminton Team

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, JavaScript

Web & Backend Technologies: HTML, CSS, JavaScript, React, Node.js, Express.js, Flask, REST APIs

Data Science & ML: Data Preprocessing, EDA, Supervised Learning (Regression & Classification), Feature Engineering, Model Training & Evaluation, NLP, Computer Vision

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, Matplotlib, LangChain, OpenCV, TensorFlow/Keras, spaCy, SpringBoot

Databases: MySQL, MongoDB, ChromaDB (Vector DB)

Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab, Postman, Ollama

Core Concepts: Object-Oriented Programming, Data Structures & Algorithms, Software Engineering Fundamentals, RAG Architecture

CONTACT

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GitHub: Rosheni-a11y

SOFT SKILLS

- Strong analytical & problem-solving abilities
- Clear communication of technical skills to non-technical audiences
- Effective team collaboration & cross-functional coordination
- Good time management with the ability to meet deadlines
- Adaptable mindset with commitment to continuous learning
- High attention to detail with focus on accuracy & quality
- Sound communication skills

CERTIFICATIONS

ML Specialization

Supervised ML: Regression & Classification
Coursera | Andrew Ng • 2026

GenAI 101 with Pieces

Generative AI, Prompt Engineering, LLM Applications • 2025

Postman API Fundamentals Student Expert

REST API Testing & Validation • 2025

Programming in Python

University of Moratuwa • 2025

LEADERSHIP & VOLUNTEERING

LeadSpring 2025

IEEE IAS & PES, SLIIT Student Branch
Publicity Team Member (Emerging Member)

Drove digital outreach campaigns for IEEE LeadSpring 2025, increasing event awareness across the student community. Collaborated with a multidisciplinary team under tight deadlines to deliver timely promotional content.

PROJECTS

1. StudyBuddy AI — RAG-Powered Study Assistant | [GitHub](#)

In Progress

Tech: Python, Flask, React, LangChain, ChromaDB, Ollama (Llama 3.1)

- Engineered a full-stack AI study assistant using Retrieval-Augmented Generation, enabling students to upload lecture PDFs and receive instant, context-aware answers — reducing study material search time by an estimated 70%.
- Built a document ingestion pipeline that chunks PDFs into 500-character segments with 50-character overlap, generates vector embeddings, and stores them in ChromaDB for semantic similarity search.
- Integrated a locally hosted Llama 3.1 LLM via Ollama, eliminating API costs while maintaining full data privacy for student materials.
- Designed a React chat interface with source attribution, tracing every AI-generated answer back to the exact PDF and page it was derived from.

2. Real-Time Emotion Detection System | [GitHub](#)

In Progress

Tech: Python, OpenCV, TensorFlow/Keras, CNN, NumPy

- Built a webcam-based application that detects and classifies 7 facial emotions (happy, sad, angry, surprise, fear, disgust, neutral) in real-time at 15+ FPS using a pre-trained CNN model.
- Leveraged transfer learning on the FER-2013 dataset (35,000+ labelled images), achieving reliable emotion classification without training a model from scratch.
- Implemented OpenCV Haar Cascade face detection with region-of-interest extraction, processing each video frame in under 60ms for smooth real-time performance.

3. Jewelry E-Commerce with AI Sales Prediction | [GitHub](#)

Group Project

Tech: MongoDB, Express.js, React, Node.js (MERN), Python, Scikit-learn | Deployed on Vercel

- Led the product management module in a 6-member team, building full CRUD operations for the jewelry catalog with image uploads, categorization, search functionality, and inventory tracking.
- Developed an AI-powered sales prediction feature using Random Forest regression trained on historical transaction data, enabling store owners to forecast monthly revenue trends with actionable insights.
- Integrated the product module with 5 other modules (custom design, design portfolio, feedback, discount, and order management) through RESTful APIs connecting the React frontend to Node.js/Express and MongoDB, with role-based authentication for admin and customer access.

4. AI-Powered Resume Analyzer | [GitHub](#)

In Progress

Tech: Python, Flask, React, NLP (spaCy/NLTK), Scikit-learn

- Building a web application that scores resume-to-job-description alignment using TF-IDF vectorization and cosine similarity, automatically identifying missing keywords and skill gaps.
- Implementing a PDF/DOCX text extraction pipeline with spaCy named entity recognition to parse education, skills, and experience sections -reducing manual resume review from 30+ minutes to under 2 minutes.
- Designing a React dashboard that visualizes match scores and generates tailored improvement suggestions to help job seekers increase interview callback rates.

Referee:

Dr Trevor Mendis (Ph.D.)

Senior Management Consultant

Member of the Board of Study

Company Director (PLC)

Postgraduate Institute of Management

University of Sri Jayewardenepura

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